
BEYOND DIFFERENCES: DOUBLY ROBUST META-LEARNERS FOR RATIO-BASED TREATMENT EFFECTS

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ABSTRACT

When treatment effects are naturally expressed as ratios — as in medicine, pricing, and marketing — the ratio-based CATE $\tau(x) = \mathbb{E}[Y|W=1, X=x]/\mathbb{E}[Y|W=0, X=x]$ is the appropriate estimand. Yet existing estimators either impose log-linear parametric structure or apply generic regression without robustness guarantees for this functional. We introduce the *Q-Learner*, which decomposes $\tau(x)$ into a product of two odds ratios, reducing ratio-CATE estimation for binary outcomes to two propensity classification tasks. We further derive doubly robust augmentations for both S/T- and Q-style ratio learners and characterize their distinct robustness properties. In benchmarks on seven RCT datasets, the Q-Learner is the most consistently competitive method in low-conversion regimes, where its propensity-only construction sidesteps the imbalanced regression that hurts outcome-based estimators. On four observational datasets, where propensity must be estimated and confounding cannot be ruled out, the DR learners introduced here decisively come out on top, making them practitioners’ natural default for confounded observational data.

Keywords Conditional Average Treatment Effect · Ratio-based CATE · Meta-Learner · Doubly Robust Estimation · Heterogeneous Treatment Effects · Uplift Modeling

1 Introduction

In many domains, treatment effects are naturally expressed as ratios. In medicine, relative effects are the dominant reporting format for both trials and subgroup analyses (Andersen, 2021). In pricing, treatment effects are most naturally expressed as elasticities — ratios of purchase rates at different price levels — rather than absolute changes in conversion. In marketing, treatment effects are routinely reported as relative lift over baseline conversion. In each case, the ratio-based CATE

$$\tau(x) = \frac{\mathbb{E}[Y|W=1, X=x]}{\mathbb{E}[Y|W=0, X=x]} \quad (1)$$

is the appropriate estimand.

The ratio scale is also often more stable than the absolute scale. Schmid et al. (1998) give empirical evidence: control-group event rates predict absolute but not relative treatment effects across clinical trials, suggesting that the ratio scale better captures the intrinsic treatment effect. The same point is visible mechanically: the same absolute effect $\tau_{\text{diff}}(x) = 0.05$ corresponds to a sixfold increase when $\mu_0(x) = 0.01$ but only a 25% lift when $\mu_0(x) = 0.2$. This mechanism is typical when baselines are low and variable — the standard regime in clinical trials (1–20%) and marketing campaigns (1–10%). When baselines are instead high and homogeneous, the difference-based CATE is equally natural and often easier to estimate.

Difference-based CATE estimation

$$\tau_{\text{diff}}(x) = \mathbb{E}[Y|W=1, X=x] - \mathbb{E}[Y|W=0, X=x] \quad (2)$$

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is a mature area, with established meta-learners (Künzel et al., 2019; Nie and Wager, 2021) and doubly robust constructions (Kennedy, 2023). The literature on ratio-based CATE estimation is much thinner. Existing approaches either impose parametric log-linear structure that cannot capture complex heterogeneity (Yadlowsky et al., 2021; Zou, 2004), or apply generic outcome regression (S- or T-Learner style; Künzel et al., 2019) without exploiting the multiplicative structure of the ratio functional and without robustness guarantees against model misspecification.

One might attempt to recover $\tau(x)$ from existing difference-based estimators via $\tau(x) = 1 + \tau_{\text{diff}}(x)/\mu_0(x)$, but this requires an additional estimate of $\mu_0(x)$, and errors in both $\hat{\tau}_{\text{diff}}$ and $\hat{\mu}_0$ compound through the division—particularly when $\mu_0(x)$ is small, which is precisely the regime where the ratio-based CATE is most valuable. Alternatively, one could turn the ratio into a difference via $\log \tau(x) = \log \mu_1(x) - \log \mu_0(x)$ and apply existing difference-based methods on the log scale. However, this still requires separate log-outcome models, and robustness guarantees derived for difference-based estimators do not transfer through nonlinear transformations. Dedicated methods for the ratio functional are therefore necessary.

1.1 Contributions

For binary outcomes (e.g., conversion or claim occurrence), where $\tau(x)$ coincides with the conditional relative risk, this paper makes two contributions. First, we introduce the *Q-Learner*, which exploits a Bayes-rule identity to decompose the ratio CATE into a product of two odds ratios:

$$\tau(x) = \frac{p(x)}{1 - p(x)} \cdot \frac{1 - e(x)}{e(x)},$$

where $p(x) = P(W=1|Y=1, X=x)$ is the converter propensity and $e(x) = P(W=1|X=x)$ is the treatment propensity. This reduces ratio CATE estimation to two binary classification tasks without requiring outcome regression. At low conversion rates, this also converts a heavily imbalanced regression task into two balanced classification tasks and sidesteps the numerical instability of $\hat{\mu}_0$ in the denominator. The Q-Learner directly optimizes $\hat{\tau}(x)$ through its component classifiers, making it naturally amenable to hyperparameter tuning and model selection. As a special case we outline the *Q-Simple Learner*, which estimates τ on the converter subset only. This is not only computationally cheaper, but sometimes a must when only data on the converters is available.

Second, we derive *doubly robust augmentations* for both S/T-style and Q-style ratio learners, on both direct and log scale, using influence-function-based corrections. These augmentations eliminate first-order bias from nuisance estimation errors, providing formal robustness guarantees. The two constructions exhibit *different* robustness properties: DR-S/T achieves classical double robustness (consistency if either propensity or outcome model is correct), while DR-Q achieves only conditional robustness requiring exact propensity knowledge—a distinction with empirical consequences in observational settings.

We evaluate all methods on seven RCT and four observational datasets. In experimental settings with known propensity, we confirm that the Q-Learner’s ability to turn an imbalanced regression problem into balanced classification tasks makes it the most consistently competitive method on low-conversion datasets. On higher-conversion datasets, where this imbalance does not arise, standard ML approaches (S-Learner) remain competitive. On observational data, where propensity must be estimated and confounding cannot be ruled out, our DR augmentations win on three of four datasets and place a strong second on the fourth, confirming the importance of double robustness when propensity is uncertain. Across both regimes, the S-Learner is the best-calibrated method.

1.2 Related Work

Meta-Learners and Doubly Robust Foundations. Difference-based CATE methodology relies on the meta-learner paradigm of Künzel et al. (2019) and Nie and Wager (2021), and on the pseudo-outcome and influence-function machinery developed in Robins et al. (1994), Chernozhukov et al. (2018), Kennedy (2023), and Foster and Syrgkanis (2023). Our DR-S/T and DR-Q constructions adapt this machinery to the ratio functional.

Ratio-based CATE Estimation. Closest to our setting, Yadlowsky et al. (2021) propose a doubly robust estimator for the ratio-based CATE restricted to a log-linear semiparametric model $D(z) = \exp(\beta_0^\top z)$. Our work imposes no parametric structure on $\tau(x)$, introduces the Q-Learner parameterisation that requires no outcome regression, and derives two distinct DR augmentations with differing robustness properties. Causal forests (Athey et al., 2019) estimate heterogeneous effects via local moment conditions and can in principle target non-standard functionals, but do not provide an out-of-the-box estimator or efficiency theory for the ratio CATE.

Relative Risk Estimation. Log-binomial and modified Poisson regression (Zou, 2004) estimate marginal or conditionally linear relative risks but cannot capture nonlinear heterogeneity. Richardson et al. (2017) develop a semiparametric

framework for the joint modelling of the relative risk and the risk difference, characterising the nuisance tangent space for these functionals. Shirvaikar et al. (2023) modify causal forests to target relative risk via GLM-based node splitting; their approach retains a GLM-based splitting criterion and does not provide doubly robust guarantees. Boughdiri et al. (2025) derive doubly robust estimators for the marginal risk ratio, establishing asymptotic normality and confidence intervals; their work targets population-level effects rather than heterogeneous CATEs. Concurrently, Gao and Hastie (2025) propose the DINA estimator (difference in natural parameters) via an R-Learner-style extension for exponential families. For Bernoulli outcomes DINA targets the conditional log-odds ratio; for Poisson (count) outcomes it coincides with the conditional log risk ratio, paralleling our log-scale variants. Their method requires choosing an outcome family, whereas the Q-Learner identity and our DR augmentations target the risk ratio directly without such a specification. DINA also differs in construction: it adapts the R-Learner’s Robinson-style loss via exponential-family likelihoods, whereas our DR learners use explicit influence-function-based pseudo-outcomes.

Uplift Modeling. The marketing literature (Radcliffe and Surry, 2011) focuses on ranking by treatment effect. We address both ranking and calibration.

The remainder of this paper is organized as follows. Section 2 establishes notation and assumptions. Section 3 derives the Q-Learner identity. Section 4 develops the doubly robust extensions. Section 5 presents the empirical evaluation, and Section 6 concludes.

2 Setup and Notation

We work within the standard potential outcomes framework (Rubin, 1974). Each unit is characterized by a vector of pre-treatment covariates $X \in \mathcal{X}$, a binary treatment indicator $W \in \{0, 1\}$, and a binary outcome $Y \in \{0, 1\}$.

Our target estimand is the ratio-based conditional average treatment effect:

$$\tau(x) = \frac{\mu_1(x)}{\mu_0(x)}, \quad (3)$$

where $\mu_w(x) = P(Y=1 \mid W=w, X=x)$ denotes the response surface under treatment arm w . The ratio $\tau(x)$ quantifies how much more likely the outcome becomes under treatment relative to control, for units with covariates x .

Identification of $\tau(x)$ from observational data requires two standard assumptions:

Assumption 1 (Unconfoundedness) $(Y(1), Y(0)) \perp W \mid X$

Assumption 2 (Overlap) $0 < e(x) < 1$ for all $x \in \mathcal{X}$,

where $e(x) = P(W=1 \mid X=x)$ is the propensity score. In addition, the ratio functional requires that the denominator is bounded away from zero:

Assumption 3 (Positivity of Control Response) $\mu_0(x) > 0$ for all $x \in \mathcal{X}$.

This assumption ensures that $\tau(x)$ is well-defined and is satisfied whenever the outcome has non-negligible baseline prevalence—the typical case in the applications we consider (conversion rates $\geq 1\%$). In practice, regions where $\mu_0(x)$ is very small lead to unstable estimates; we address this through clipping (Appendix B).

Beyond these standard quantities, the Q-Learner (Section 3) relies on two additional objects:

- The **converter propensity** $p(x) = P(W=1 \mid Y=1, X=x)$: the probability of having been treated, conditional on a positive outcome.
- The **marginal conversion probability** $m(x) = P(Y=1 \mid X=x)$: the unconditional outcome rate given covariates.

3 The Q-Learner

We now derive a meta-learner that directly exploits the multiplicative structure of the ratio CATE. The key observation is the following identification result:

Under Assumptions 1–3, for binary outcomes $Y \in \{0, 1\}$:

$$\tau(x) = \frac{p(x)}{1 - p(x)} \cdot \frac{1 - e(x)}{e(x)} \quad (4)$$

where $p(x) = P(W=1|Y=1, X=x)$ is the treatment probability among converters and $e(x) = P(W=1|X=x)$ is the propensity. To prove the relation, apply Bayes' theorem to $\mu_1(x) = P(Y=1|W=1, X=x)$:

$$\mu_1(x) = \frac{P(W=1|Y=1, X=x) \cdot P(Y=1|X=x)}{P(W=1|X=x)} = \frac{p(x) \cdot m(x)}{e(x)} \quad (5)$$

Similarly, $\mu_0(x) = (1 - p(x)) \cdot m(x)/(1 - e(x))$. Taking the ratio, $m(x)$ cancels:

$$\tau(x) = \frac{\mu_1(x)}{\mu_0(x)} = \frac{p(x)}{1 - p(x)} \cdot \frac{1 - e(x)}{e(x)} \quad (6)$$

The identity has an intuitive interpretation: if treatment increases conversion ($\tau(x) > 1$), then converters are disproportionately likely to have been treated relative to the overall propensity, i.e., $p(x) > e(x)$. The ratio $\frac{p(x)/(1-p(x))}{e(x)/(1-e(x))}$ quantifies exactly this excess on the odds scale.

The Q-Learner identity suggests a simple estimation strategy:

1. **Step 1:** Estimate propensity score $\hat{e}(x)$ from the observed treatment assignments using all observations (Section 3.1 explains why estimation is preferred even when $e(x)$ is known)
2. **Step 2:** Estimate converter treatment probability $\hat{p}(x)$ using only observations with $Y = 1$
3. **Step 3:** Compute CATE estimate: $\hat{\tau}(x) = \frac{\hat{p}(x)}{1 - \hat{p}(x)} \cdot \frac{1 - \hat{e}(x)}{\hat{e}(x)}$

Compared to S- and T-Learners, the Q-Learner offers three structural advantages:

- **Direct optimization of τ :** Since $\hat{\tau}(x)$ is a deterministic function of $\hat{p}(x)$ and $\hat{e}(x)$, hyperparameter tuning of the component models directly improves the treatment effect estimate. In contrast, the S-Learner optimizes the outcome model $\hat{\mu}(x, w)$, and improvements in outcome prediction do not necessarily translate to better treatment effect estimation. This makes the Q-Learner particularly amenable to automated model selection (Section 5).
- **Balanced classification:** In A/B tests with $e(x) \approx 0.5$ and low conversion rates, the Q-Learner transforms a highly imbalanced outcome regression problem into a balanced classification problem among converters. We expect this to make the Q-Learner especially competitive on low-conversion datasets, an expectation we confirm empirically in Section 5.
- **Flexibility:** Any probabilistic classifier can serve as base model for both stages.

Remark 1 (Single Model via Offset) Taking logs of (4) one arrives at

$$\log \tau(x) = \text{logit}(p(x)) - \text{logit}(e(x)). \quad (7)$$

This suggests fitting a single logistic regression for $p(x)$ with $\text{logit}(\hat{e}(x))$ as offset:

$$\text{logit}(p(x)) = \text{logit}(\hat{e}(x)) + f(x) \quad (8)$$

where $f(x)$ models the treatment effect heterogeneity. The fitted model directly yields $\hat{\tau}(x) = \exp(\hat{f}(x))$, reducing the two-stage procedure to a single model after propensity estimation.

Remark 2 (Extension to Multi-Class Treatments) The Q-Learner can be generalized to non-binary treatments by using multi-class classification for $p(w, x) = P(W=w|Y=1, X=x)$ and $e(w, x) = P(W=w|X=x)$:

$$\tau(x, w_1, w_2) = \frac{p(w_1, x)}{p(w_2, x)} \cdot \frac{e(w_2, x)}{e(w_1, x)} \quad (9)$$

Extensions to non-binary outcomes are possible but less straightforward; we leave their investigation to future work.

3.1 Propensity Estimation and the Q-Simple Learner

In RCT settings where propensity is exactly known, one might be tempted to skip fitting a propensity model and use the known $e(x)$ directly in the Q-Learner identity (4). This reduces CATE estimation to a single model for $p(x)$, estimated only among converters — a computationally attractive simplification, especially for large datasets. We call this variant the *Q-Simple Learner*.

Beyond its computational appeal, Q-Simple has a categorical advantage that no other learner in the family shares: it is the only one that can be applied when only converters are observed, since $p(x)$ is fitted on the $Y = 1$ subset and the design propensity replaces the full-sample model. This is the natural data layout in operational settings such as price-elasticity estimation on existing customers, where covariates and outcomes are collected only post-conversion.

Estimating propensity from data is nonetheless preferable when possible, because it can reduce finite-sample variance. The key insight is that $\hat{p}(x)$ and $\hat{e}(x)$ are correlated through the same realized treatment assignments: if a covariate region happens to contain more treated units than expected, both $\hat{e}(x)$ and $\hat{p}(x)$ increase, and the biases partially cancel in the product $\frac{\hat{p}}{1-\hat{p}} \cdot \frac{1-\hat{e}}{\hat{e}}$. Using the known $e(x)$ breaks this correlation, potentially increasing finite-sample variance. Fitting $\hat{e}(x)$ from data—even though the model only captures the randomness of the treatment assignments—preserves this correlation and thereby reduces variance. The reduction diminishes with sample size: as n grows, the realized propensity concentrates around its true value, and the correlation benefit vanishes, so for large datasets the computational savings from skipping propensity estimation may outweigh the small variance cost.

Section 5 confirms that the Q-Learner outperforms Q-Simple, although on most datasets the gap is small (see Figure 1). For applications where computational efficiency matters or where only converter data is available, Q-Simple therefore remains a competitive choice on most RCTs.

4 Doubly Robust Extensions

The Q-Learner and the naive ratio estimator $\hat{\mu}_1(x)/\hat{\mu}_0(x)$ (T-Learner) are plug-in estimators: they combine estimated nuisance functions into $\hat{\tau}(x)$ without correction for estimation error. If any nuisance model is misspecified, bias propagates directly into the CATE estimate.

Following the framework of Kennedy (2022), we derive *augmented* estimators that eliminate the leading-order bias through influence-function-based corrections. Concretely, for an estimand $\tau(x)$ that depends on nuisance functions g (propensity, outcome models, etc.), the augmented pseudo-outcome takes the form

$$\Gamma = \hat{\tau}(x) + \hat{\phi}_\tau(W, Y, X; x), \quad (10)$$

where $\hat{\phi}_\tau$ is the estimated influence function of τ evaluated at the nuisance estimates. This correction removes the first-order (linear) bias term, leaving a remainder of order $O_p(\|\hat{g} - g\|_2^2)$ —a product of nuisance estimation errors that vanishes faster than any individual error alone. The resulting estimators are *oracle-efficient* in the sense of Kennedy (2022) when nuisance estimates converge at sufficiently fast rates: regressing the pseudo-outcome Γ on X then achieves the same convergence rate as if the true nuisance functions were known.

The ratio CATE $\tau(x) = \mu_1(x)/\mu_0(x)$ admits two natural parameterizations: through outcome models (μ_1, μ_0) or through the Q-Learner reparameterization (p, e) . Each leads to a distinct augmented estimator. As we show below, these exhibit fundamentally different robustness properties: DR-S/T achieves classical double robustness, while DR-Q achieves only conditional robustness requiring exact propensity knowledge.

4.1 DR-S/T: Augmenting the Ratio via Outcome Models

Both estimate $\hat{\tau}(x) = \hat{\mu}_1(x)/\hat{\mu}_0(x)$ as a plug-in ratio and share the following augmentation. The standard AIPW influence functions for the response surfaces are:

$$\phi_{\mu_1}(W, Y, X; x) = \frac{W(Y - \mu_1(x))}{e(x)}, \quad (11)$$

$$\phi_{\mu_0}(W, Y, X; x) = \frac{(1 - W)(Y - \mu_0(x))}{1 - e(x)}. \quad (12)$$

Applying the quotient rule $\phi_\tau = (\phi_{\mu_1} - \tau \cdot \phi_{\mu_0})/\mu_0$ and substituting nuisance estimates yields the DR-S/T pseudo-outcome:

$$\Gamma^{\text{DR-S/T}} = \hat{\tau}(X) + \frac{W(Y - \hat{\mu}_1(X))}{\hat{e}(X) \cdot \hat{\mu}_0(X)} - \frac{\hat{\tau}(X)(1 - W)(Y - \hat{\mu}_0(X))}{(1 - \hat{e}(X)) \cdot \hat{\mu}_0(X)} \quad (13)$$

where $\hat{\tau}(X) = \hat{\mu}_1(X)/\hat{\mu}_0(X)$. The first correction term adjusts for bias in $\hat{\mu}_1$, the second for bias in $\hat{\mu}_0$, both weighted by the inverse propensity.

4.1.1 Double Robustness of DR-S/T

The DR-S/T estimator achieves *classical double robustness*: it is consistent for $\tau(x)$ if either

- (a) the propensity model is correct: $\hat{e}(x) = e(x)$, or
- (b) the outcome models are correct: $\hat{\mu}_0(x) = \mu_0(x)$ and $\hat{\mu}_1(x) = \mu_1(x)$.

We now verify both cases. Writing $\Gamma^{\text{DR-S/T}} = A + B - C$ with $A = \hat{\tau}$, $B = W(Y - \hat{\mu}_1)/(\hat{e}\hat{\mu}_0)$, and $C = \hat{\tau}(1 - W)(Y - \hat{\mu}_0)/((1 - \hat{e})\hat{\mu}_0)$:

- **Case (a): $\hat{e} = e$ (correct propensity).**

Taking conditional expectations given $X = x$:

$$\mathbb{E}[B \mid X=x] = \frac{\mu_1(x) - \hat{\mu}_1(x)}{\hat{\mu}_0(x)}, \quad (14)$$

$$\mathbb{E}[C \mid X=x] = \frac{\hat{\tau}(x)(\mu_0(x) - \hat{\mu}_0(x))}{\hat{\mu}_0(x)}. \quad (15)$$

A first-order Taylor expansion of $\tau = \mu_1/\mu_0$ around $(\hat{\mu}_1, \hat{\mu}_0)$ gives:

$$\tau(x) = \hat{\tau}(x) + \frac{\mu_1(x) - \hat{\mu}_1(x)}{\hat{\mu}_0(x)} - \frac{\hat{\tau}(x)(\mu_0(x) - \hat{\mu}_0(x))}{\hat{\mu}_0(x)} + O_p(\|\hat{\mu} - \mu\|_2^2) \quad (16)$$

Therefore $\mathbb{E}[\Gamma^{\text{DR-S/T}} \mid X=x] = \tau(x) + O_p(\|\hat{\mu} - \mu\|_2^2)$.

- **Case (b): $\hat{\mu}_0 = \mu_0, \hat{\mu}_1 = \mu_1$ (correct outcome models).**

Then $A = \tau(x)$, and $\mathbb{E}[B \mid X=x] = \mathbb{E}[C \mid X=x] = 0$ since $Y - \mu_w(X)$ has conditional mean zero given (W, X) . Thus $\mathbb{E}[\Gamma^{\text{DR-S/T}} \mid X=x] = \tau(x)$ exactly, regardless of the propensity estimate.

When both models are misspecified, the bias is $O_p(\|\hat{e} - e\|_2 \cdot \|\hat{\mu} - \mu\|_2)$, second-order in the nuisance estimation errors.

4.1.2 DR-S/T on Log Scale

When treatment effects span a wide range, direct-scale pseudo-outcomes exhibit heteroscedastic variance proportional to the effect magnitude. A log-scale formulation provides variance stabilization. Since $\log \tau(x) = \log \mu_1(x) - \log \mu_0(x)$ and the influence functions for $\log \mu_w$ are $\phi_{\log \mu_w} = \phi_{\mu_w}/\mu_w$, we obtain:

$$\Gamma_{\log}^{\text{DR-S/T}} = \log \hat{\mu}_1(X) - \log \hat{\mu}_0(X) + \frac{W(Y - \hat{\mu}_1(X))}{\hat{e}(X)\hat{\mu}_1(X)} - \frac{(1 - W)(Y - \hat{\mu}_0(X))}{(1 - \hat{e}(X))\hat{\mu}_0(X)} \quad (17)$$

The log-scale DR-S/T inherits the same double robustness property as the direct-scale version.

4.2 DR-Q: Augmenting the Q-Learner

While DR-S/T achieves classical double robustness, the Q-Learner parameterization leads to a weaker robustness structure. The Q-Learner identity (4) expresses $\tau(x)$ as a product:

$$\tau(x) = \underbrace{\frac{p(x)}{1 - p(x)}}_{A(x)} \cdot \underbrace{\frac{1 - e(x)}{e(x)}}_{B(x)} \quad (18)$$

We apply the product rule $\phi_\tau = B\phi_A + A\phi_B$. For $A(x) = p(x)/(1 - p(x))$, the influence function on the converter subpopulation, lifted to the full population via $m(x)$, is $\phi_A = Y(W - p(x))/(m(x)(1 - p(x))^2)$. For $B(x) = (1 - e(x))/e(x)$, the influence function is $\phi_B = -(W - e(x))/e(x)^2$. Substituting nuisance estimates yields the DR-Q pseudo-outcome:

$$\Gamma^{\text{DR-Q}} = \hat{\tau}(X) + \frac{(1 - \hat{e}(X)) \cdot Y(W - \hat{p}(X))}{\hat{e}(X) \cdot \hat{m}(X)(1 - \hat{p}(X))^2} - \frac{\hat{p}(X)(W - \hat{e}(X))}{(1 - \hat{p}(X))\hat{e}(X)^2} \quad (19)$$

where $\hat{\tau}(X) = \frac{\hat{p}(X)}{1 - \hat{p}(X)} \cdot \frac{1 - \hat{e}(X)}{\hat{e}(X)}$ is the plug-in Q-Learner estimate. Since both parameterizations target the same functional $\tau(x)$, the augmentation terms are algebraically equivalent to those of DR-S/T (13) after substitution of the identities $\mu_1 = pm/e$ and $\mu_0 = (1 - p)m/(1 - e)$.

4.2.1 Conditional Robustness of DR-Q

Unlike DR-S/T, the DR-Q estimator does *not* achieve classical double robustness: the propensity score $e(x)$ appears in *both* the plug-in estimate and the augmentation terms.

Concretely, the DR-Q estimator requires *exact* specification of the propensity score $e(x)$. Given $\hat{e}(x) = e(x)$ **exactly**, the estimator is consistent for $\tau(x)$ if either:

- (a) the marginal conversion model is correct: $\hat{m}(x) = m(x)$, or
- (b) the converter propensity model is correct: $\hat{p}(x) = p(x)$

If $\hat{e}(x) \neq e(x)$, the estimator is generally inconsistent regardless of the quality of $\hat{m}$ and $\hat{p}$.

We verify both cases. Writing

$$\Gamma^{\text{DR-Q}} = \underbrace{\hat{A}\hat{B}}_{\text{plug-in}} + \underbrace{\hat{B} \cdot \hat{\phi}_A}_{p\text{-correction}} + \underbrace{\hat{A} \cdot \hat{\phi}_B}_{e\text{-correction}} \quad (20)$$

where $\hat{A} = \hat{p}/(1 - \hat{p})$, $\hat{B} = (1 - \hat{e})/\hat{e}$, and $\hat{\phi}_A, \hat{\phi}_B$ are the estimated influence function contributions.

- **Case (a): $\hat{e} = e$ and $\hat{m} = m$ (correct propensity and marginal conversion).**

With correct propensity, $\hat{B} = B$ and $\mathbb{E}[\hat{\phi}_B | X] = 0$. For the p -correction term, note that $\mathbb{E}[Y(W - \hat{p}(X)) | X=x] = m(x)(p(x) - \hat{p}(x))$, so with $\hat{m} = m$ the marginal conversion cancels:

$$\mathbb{E}[\hat{B} \cdot \hat{\phi}_A | X=x] = B(x) \cdot \frac{p(x) - \hat{p}(x)}{(1 - \hat{p}(x))^2} \quad (21)$$

Using a first-order Taylor expansion of $A = p/(1 - p)$ around $\hat{p}$:

$$A(x) = \hat{A}(x) + \frac{p(x) - \hat{p}(x)}{(1 - \hat{p}(x))^2} + O_p(\|\hat{p} - p\|_2^2) \quad (22)$$

Therefore:

$$\mathbb{E}[\Gamma^{\text{DR-Q}} | X=x] = \hat{A}(x)B(x) + B(x) \frac{p(x) - \hat{p}(x)}{(1 - \hat{p}(x))^2} = \tau(x) + O_p(\|\hat{p} - p\|_2^2) \quad (23)$$

- **Case (b): $\hat{e} = e$ and $\hat{p} = p$ (correct propensity and converter propensity).**

Then $\hat{A} = A$, $\hat{B} = B$, so the plug-in equals $\tau(x)$. Both correction terms have conditional mean zero: $\mathbb{E}[\hat{\phi}_A | X] = 0$ since $W - p(X)$ has mean zero given $(Y=1, X)$, and $\mathbb{E}[\hat{\phi}_B | X] = 0$ since $W - e(X)$ has mean zero given X . Thus $\mathbb{E}[\Gamma^{\text{DR-Q}} | X=x] = \tau(x)$ exactly.

4.2.2 DR-Q on Log Scale

Taking logarithms of the Q-Learner identity gives $\log \tau(x) = \text{logit}(p(x)) - \text{logit}(e(x))$, where $\text{logit}(u) = \log(u/(1 - u))$. Applying the delta method with $\frac{d}{dp} \text{logit}(p) = 1/(p(1 - p))$:

$$\Gamma_{\log}^{\text{DR-Q}} = \underbrace{\text{logit}(\hat{p}(X)) - \text{logit}(\hat{e}(X))}_{\text{plug-in}} + \underbrace{\frac{Y(W - \hat{p}(X))}{\hat{m}(X)\hat{p}(X)(1 - \hat{p}(X))}}_{\text{correction for } p} - \underbrace{\frac{W - \hat{e}(X)}{\hat{e}(X)(1 - \hat{e}(X))}}_{\text{correction for } e} \quad (24)$$

The log-scale DR-Q has the same conditional robustness properties as the direct-scale version (Section 4.2.1).

4.2.3 DR-Q-Simple Learner

The conditional robustness of DR-Q demands $\hat{e}(x) = e(x)$. Under this requirement, the e -augmentation term has conditional expectation zero and provides no bias correction. Analogous to the Q-Simple Learner (Section 3.1) this motivates a simplified construction that operates only on converters ($Y = 1$) and eliminates the need to estimate $m(x)$:

$$\Gamma^{\text{DR-Q-Simple}} = \hat{\tau}(X) + \frac{(1 - e(X)) \cdot (W - \hat{p}(X))}{e(X) \cdot (1 - \hat{p}(X))^2} \quad (25)$$

where $\hat{\tau}(X) = \frac{\hat{p}(X)}{1 - \hat{p}(X)} \cdot \frac{1 - e(X)}{e(X)}$ and $e(x)$ is the known propensity. Given exact $e(x)$:

$$\mathbb{E}[\Gamma^{\text{DR-Q-Simple}} | X = x, Y = 1] = \tau(x) + O_p(\|\hat{p} - p\|_2^2) \quad (26)$$

The DR-Q-Simple learner is particularly attractive for RCT applications: it eliminates the need to estimate $m(x)$, operates only on the $n_1 = \sum_i Y_i$ converters, and requires only a single nuisance function $\hat{p}(x)$. The simplification comes at a cost in asymptotic efficiency.

4.2.4 DR-Q-Simple on Log Scale

Applying the same log-scale construction as in DR-Q (Section 4.1.2) but restricting to converters and using known $e(x)$:

$$\Gamma_{\log}^{\text{DR-Q-Simple}} = \underbrace{\text{logit}(\hat{p}(X)) - \text{logit}(e(X))}_{\text{plug-in}} + \underbrace{\frac{W - \hat{p}(X)}{\hat{p}(X)(1 - \hat{p}(X))}}_{\text{correction for } p} \quad (27)$$

This pseudo-outcome is computed only for observations with $Y = 1$ and inherits the same robustness property: consistency for $\log \tau(x)$ with bias $O_p(\|\hat{p} - p\|_2^2)$.

5 Empirical Evaluation

We now systematically evaluate which learner performs best under which conditions. While the limited number of available datasets makes definitive conclusions difficult, the empirical patterns — backed by the theoretical analysis in Sections 3–4 — point to two factors that govern estimator choice: the **baseline conversion rate** and **whether the data is experimental or observational**. Section 5.1 describes the benchmark protocol; Section 5.2 reports the ranking results; Section 5.3 reports calibration; and Section 5.4 concludes with practical recommendations.

5.1 Setup

Datasets. We select datasets to cover a range of sample sizes ($n = 1.6\text{k}–13.9\text{M}$), conversion rates (0.9%–68%), and treatment fractions (18%–85%). We separate RCT datasets, where propensity is known by design, from observational datasets, where propensity must be estimated and confounding is present. This separation reflects the fundamentally different estimation challenges and allows us to test the theoretical distinction between classical and conditional double robustness.

Table 1: RCT benchmark datasets. Propensity $e(x)$ is known by design. *Semi-synthetic: both potential outcomes observed, treatment simulated with $e = 0.5$.

Dataset	n	Conv.	Treat.	Seeds	Domain
Criteo (Diemert et al., 2018)	13.9M	4.7%	85.0%	50	E-commerce
Hillstrom (Visit) (Hillstrom, 2008)	64k	14.7%	66.8%	50	Email
Hillstrom (Conv.) (Hillstrom, 2008)	64k	0.9%	66.8%	550	Email
MegaFon (MegaFon, 2019)	600k	20.4%	50.1%	50	Telecom
X5 Retail (X5 Retail, 2020)	200k	62.0%	49.9%	350	Retail
Lenta (Lenta, 2021)	687k	10.8%	75.1%	350	Grocery
Twins* (Louizos et al., 2017)	71k	3.6%	50.0%	550	Neonatal

Table 2: Observational benchmark datasets. Propensity $e(x)$ must be estimated.

Dataset	n	Conv.	Treat.	Seeds	Domain
RHC (Connors et al., 1996)	5.7k	64.9%	38.1%	550	Medical
Cattaneo (Cattaneo, 2010)	4.6k	6.0%	18.6%	350	Birthweight
NHEFS (Hernán and Robins, 2020)	1.6k	19.5%	26.3%	150	Smoking
JTPA (Abadie et al., 2002)	9.2k	50.0%	44.0%	150	Job training

Learners. The benchmark covers three categories. *Standard plug-in baselines*: the S-Learner (standard ML with treatment as a feature) and T-Learner of Künzel et al. (2019), applied to ratio CATE as $\hat{\tau} = \hat{\mu}_1 / \hat{\mu}_0$. *Our contributions*: the Q-Learner and Q-Simple Learner introduced in Section 3, and the doubly robust augmentations DR-S, DR-T, DR-Q, and DR-Q-Simple from Section 4, each in direct- and log-scale form. Q-Simple and DR-Q-Simple are only applicable to RCT datasets where the propensity is known by design. *Difference-CATE comparators*: the X-Learner of Künzel et al. (2019), the R-Learner of Nie and Wager (2021), and a classical AIPW DR-Learner; we recover ratio CATE from each via $\hat{\tau}_{\text{ratio}} = 1 + \hat{\tau}_{\text{diff}} / \hat{\mu}_0$.

Metrics. We evaluate two complementary metrics. The *Qini coefficient* measures ranking quality: how well a learner identifies individuals with the highest treatment effect. We adapt the normalized Qini coefficient (Radcliffe and Surry,

2011) to ratio-based effects by integrating the ratio rather than the difference:

$$\text{Qini}_l(d) = \int_0^1 \frac{\hat{Y}^{(1)}(s)}{\hat{Y}^{(0)}(s)} ds - \frac{Y^{(1)}}{Y^{(0)}},$$

where $\hat{Y}^{(w)}(s)$ is the mean outcome in arm w among the top- s fraction ranked by predicted $\hat{\tau}(x)$ from learner l on dataset d . Higher is better.

The *multiplicative calibration error* measures how accurately predicted effect sizes match realized effects. We compute the error on the log scale, where over- and under-prediction are treated symmetrically ($\log(0.8) = -\log(1.25)$), then exponentiate to obtain an interpretable multiplicative factor:

$$\text{CalError} = \exp \left(\frac{\sum_{b=1}^B n_b |\log \bar{\tau}_b - \log \hat{\tau}_b^{\text{emp}}|}{\sum_{b=1}^B n_b} \right), \quad (28)$$

where $\bar{\tau}_b$ is the mean predicted effect in bin b and $\hat{\tau}_b^{\text{emp}} = \bar{Y}_b^{(1)} / \bar{Y}_b^{(0)}$ is the empirical ratio within that bin ($B = 10$ equal-frequency bins). A CalError of $1.0\times$ indicates perfect calibration; $1.5\times$ means predictions are on average off by a factor of 1.5.

We regard Qini as more important than calibration: ranking failures cannot be undone, while miscalibration is recoverable. A preliminary investigation suggests the calibration gap from Q-style learners is well captured by a simple linear transformation, with the recalibration parameters stable across base learners (LightGBM, XGBoost, CatBoost) and across temporal splits — suggesting that the miscalibration is a structural property of the propensity-classifier training rather than dataset-specific noise. A thorough analysis of post-hoc recalibration is beyond this paper’s scope; we report raw CalError values throughout.

Comparing learner performance. We judge every learner by its capability to outperform standard ML (the S-Learner) — only then is the meta-learner’s existence justified. For this reason we evaluate

$$R_l(d) = \frac{\text{Qini}_l(d)}{\text{Qini}_S(d)}, \quad (29)$$

with $R_l(d) > 1$ indicating that the meta-learner beats standard ML on dataset d and $R_l(d) < 1$ indicating that it has added complexity without a gain. Whether a learner’s $R_l(d)$ is statistically distinguishable from 1 is determined by a paired test on per-seed differences $\delta_l(\text{seed}) = \text{Qini}_l(\text{seed}) - \text{Qini}_S(\text{seed})$, with significance threshold $z = 1.96$. Pairing exploits the fact that all learners share the same train/test splits per seed, removing seed-level common variance and providing substantially more power than an unpaired comparison. Per-dataset seed counts (Tables 1 and 2) vary because we applied more seeds to noisier datasets, where this paired test failed to reach significance at lower counts. Since $R_l(d)$ values vary heavily across datasets, we refrain from cross-dataset aggregation: any aggregate would be dominated by a small number of datasets. Calibration is reported on raw scale (Section 5.3); since the S-Learner dominates calibration on every dataset, no normalization is needed there. Full per-dataset Qini values and standard errors are in Appendix A.

Implementation. All methods use LightGBM (Ke et al., 2017) for nuisance and final-stage models with default hyperparameters. DR learners use 5-fold cross-fitting (Chernozhukov et al., 2018) to generate out-of-fold nuisance predictions. Final-stage regressors use a Poisson objective to enforce $\hat{\tau}(x) > 0$. To improve stability, we clip nuisance estimates and pseudo-outcomes as described in Appendix B. Each seed determines a fresh train/test split.

5.2 Ranking Analysis

We evaluate which learners most consistently achieve near-optimal ranking performance, measured by $R_l(d) = \text{Qini}_l(d) / \text{Qini}_S(d)$ relative to the S-Learner baseline. The two data regimes behave differently enough that we present them separately: RCTs in Figure 1 (sorted by conversion rate) and observational datasets in Figure 2. Full per-dataset Qini values and heatmaps are provided in Appendix A.

5.2.1 RCT Datasets

On RCT data the best learner varies heavily across datasets, so for optimal results all candidates should be checked. Still, clear patterns emerge from Figure 1 (full per-dataset values and a heatmap in Appendix A) that suggest standard procedures for the practitioner.

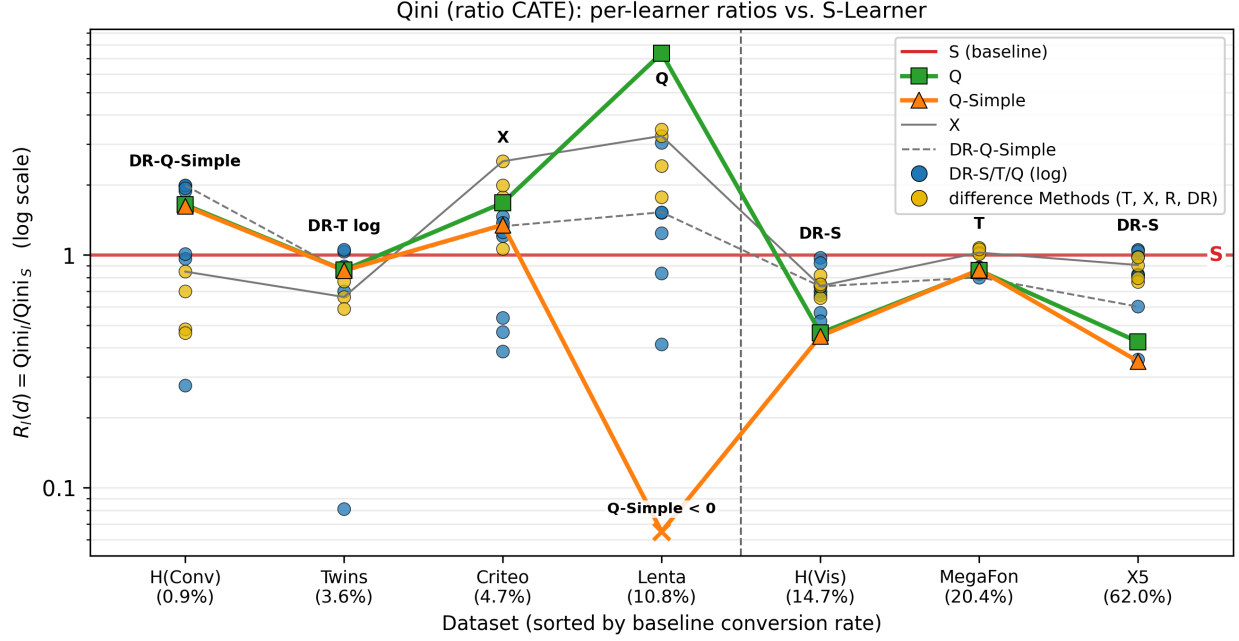


Figure 1: $Qini_l/Qini_S$ across the seven RCT datasets, sorted by conversion rate (low to high). Q-Learner and Q-Simple are highlighted as coloured lines; the remaining non-baseline learners appear as a dot cloud, grouped by family. DR-Q-Simple and X are overlaid as thin grey lines for context. The dashed vertical line separates the four low-conversion datasets (left) from the three high-conversion ones (right). The S-Learner baseline is the red line at 1.

No structural advantage for DR. Without confounding, DR augmentation is not structurally advantageous: although a DR variant occasionally takes the top spot, no DR learner is consistently ahead of the leading plug-in. The theoretical reading is that when propensity is known by design and no confounding can arise, plug-in learners already identify the CATE, and the DR correction trades a small bias gain against added estimation variance.

High CVR: standard ML suffices. On the three high-conversion datasets (X5, MegaFon, Criteo), the S-Learner sits within a hair of the dataset winner. Causal meta-learners sometimes edge it out, but only slightly. Apparently, without the imbalance problem that motivates ratio-CATEs, an out-of-the-box GBM is enough.

Low CVR: Q is the most consistent choice. On the four low-conversion datasets (H(Conv), Twins, Criteo, Lenta), the Q-Learner significantly beats the S-Learner on three out of four. The single exception is Twins, where Q sits at 0.82 of the dataset best—a significant loss against S, but small in absolute terms. Only two other learners match Q’s win count: DR-Q-Simple (three significant wins) and X (two wins, one tie). But when comparing the actual ratios, Q is clearly ahead of both: averaged across the four low-conversion datasets, $Qini_l/Qini_{\max}$ is 0.83 for Q against 0.64 for DR-Q-Simple and 0.62 for X, reflecting the visibly larger swings of the latter two across datasets.

Q-Learner vs. Q-Simple. On six of seven RCTs Q-Simple matches the full Q-Learner closely. The single exception is Lenta, where Q-Simple’s Qini collapses to essentially zero while Q reaches 0.089. Outside of this failure mode Q-Simple remains a competitive, computationally cheaper alternative when known propensity makes it applicable.

5.2.2 Observational Datasets

On observational data, where propensity must be estimated and confounding can bias plug-in estimates, the result reverses: the DR augmentation now *earns its keep*. The new DR learners introduced in this paper consistently outperform both plug-in estimators and out-of-the-box ML baselines. The new DR family is therefore the practitioner’s go-to choice once confounding is in play.

The new methods take the top spot on three of four observational datasets, while the single loss is to the R-Learner on RHC; since the R-Learner is unremarkable on the other three observational datasets (and on the RCT panel as well), we

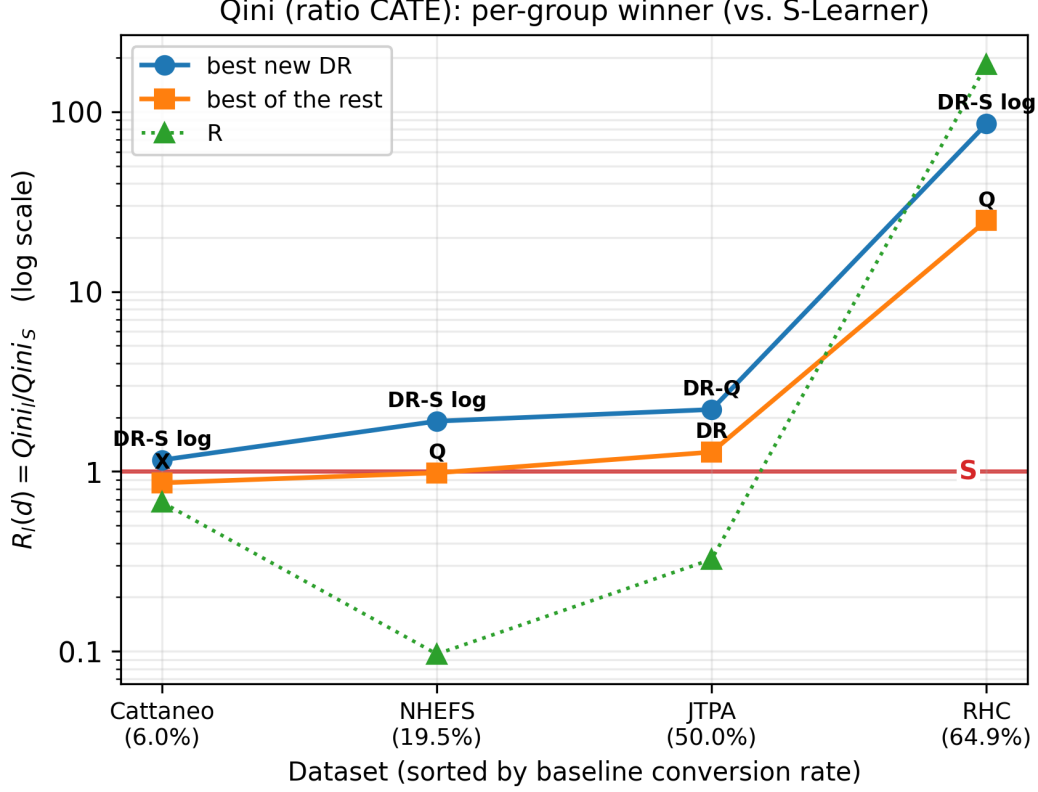


Figure 2: Best new-DR learner vs. best non-DR competitor (T, Q, Q-Simple, X, DR) on the four observational datasets, sorted by baseline conversion rate. Each marker is labelled with the winning learner of its group. The R-Learner is overlaid separately as a dotted line: it sits below the DR line on Cattaneo, NHEFS, and JTPA, and spikes far above on RHC. The S-Learner baseline ($R = 1$) is shown in red.

read this as an isolated quirk of the RHC data rather than evidence for an alternative recipe. Figure 2 summarises this picture by plotting the best new-DR learner against the best non-DR competitor for each observational dataset, with the R-Learner overlaid as a dotted line to make the RHC exception visible. The unusually high $Qini_i/Qini_S$ values on RHC reflect the fact that the S-Learner Qini drops to essentially zero—no better than random ranking—further underpinning the need for specialised bias-correction methods like the DR family on confounded data.

Direct vs. log scale. On the two low-conversion datasets (Cattaneo, NHEFS), the log-scale DR variants beat their direct-scale counterparts by a wide margin, while on the higher-conversion datasets (JTPA, RHC) the gap closes and direct-scale DR can even take the lead. The mechanism is the one that motivated the log transform in Section 4.1.2: when $\mu_0(x)$ is small, direct-scale pseudo-outcomes are inflated by a $1/\hat{\mu}_0(x)$ factor that turns small denominator errors into large pseudo-outcome swings, whereas the log transform replaces the division with a subtraction and stabilises the regression target. On the higher-conversion datasets this risk is reduced, and the direct scale’s lower bias becomes attractive again.

DR-Q vs. DR-S/T. Within the new DR family, DR-Q is rarely the outright per-dataset winner; DR-S and DR-T (or their log counterparts) usually occupy the top spot. This is consistent with the theoretical prediction from Section 5: DR-Q only achieves *conditional* robustness, while DR-S/T retain the classical double-robust guarantee, so a residual gap was expected. The cleanest read on this gap comes from the log-scale variants, which share a variance-stabilised scale and isolate the robustness comparison: DR-Q log even overtakes DR-S/T log on JTPA, ties them on NHEFS, and lags only modestly on Cattaneo and RHC. The conditional-robustness penalty is therefore mild, suggesting good propensity estimation.

5.3 Calibration Analysis

Calibration tells a one-sentence story: the **S-Learner is the best- or near-best-calibrated method on every dataset**. It wins outright on six of eleven datasets and loses by a tight margin on the other five. Direct-scale DR variants are persistently miscalibrated by factors of 2–10 $\times$ the S-Learner; log-scale DR variants, the X- and R-comparators, and the plug-in T- and Q-Learners fall in between. Per-dataset values are in Tables 5 and 6 (Appendix A). As discussed in Section 5.1, the miscalibration of Q-style learners is a structural property of the propensity-classifier training and recoverable via a simple linear post-hoc transformation; full analysis is beyond this paper’s scope.

5.4 Summary and Practical Guidance

For practitioners, the patterns observed in our benchmark suggest the following default choices:

- **RCT, high conversion rate:** S-Learner.
- **RCT, low conversion rate:** Q-Learner.
- **Observational data:** one of DR-S / DR-T / DR-Q (log-scale at low conversion, direct-scale otherwise).

Limitations. Our evaluation uses binary outcomes, a single base learner (LightGBM with default hyperparameters), and fixed clipping thresholds. Results may differ with learner-specific hyperparameter tuning, other model classes, or continuous outcomes. The observational datasets are small ($n = 1.6\text{k}–9.2\text{k}$), limiting the precision of comparisons. Without access to true CATEs, we evaluate ranking and calibration but cannot assess pointwise estimation accuracy.

6 Conclusion

We introduced the Q-Learner, a meta-learner for ratio-based CATEs that decomposes $\tau(x)$ into a product of odds ratios via a Bayes-rule identity, reducing estimation to two binary classification tasks that directly optimize the treatment effect estimate. As a special case we outlined the Q-Simple Learner, which estimates τ on the converter subset only — applicable when only converter data is available. We derived doubly robust augmentations for both outcome-regression-based (DR-S/T) and Q-Learner-based (DR-Q) ratio estimators, and characterized their distinct robustness properties: classical double robustness for DR-S/T versus conditional robustness for DR-Q.

Our empirical benchmark identifies two regimes where the methods introduced here come out on top. First, low-conversion RCTs: the Q-Learner’s reformulation of an imbalanced outcome regression as two balanced classification tasks makes it the most consistent choice in the regime where ratio-CATEs are most valuable. Second, observational data: when propensity must be estimated, a DR variant introduced in this paper takes the top spot on three of four datasets, while remaining among the leading methods on the fourth.

Ultimately, these results highlight a structural limitation in applying existing CATE estimators—in particular difference-based methods—to ratio problems. Division by baseline estimates compounds nuisance errors, inflates variance in low-baseline regions, and breaks doubly robust guarantees. The methods introduced here—the Q-Learner’s factorization into classification tasks for low-conversion experiments, and the DR augmentations for observational settings—demonstrate that targeting the ratio functional directly is the right answer in both regimes the literature has so far underserved.

Several avenues for future work remain. The influence functions established here provide a natural foundation for developing robust confidence intervals for ratio-based CATEs, which is essential for formal inference. Furthermore, the underlying framework invites natural extensions to continuous and time-to-event outcomes. Finally, evaluating the impact of learner-specific hyperparameter tuning on the Q-Learner is an important practical next step; given that its component classifiers directly optimize the $\hat{\tau}(x)$ estimand, customized tuning likely offers greater performance leverage than it would for standard difference-based meta-learners.

Code and data availability. All code, dataset loaders, and scripts to reproduce the experiments in this paper are available at <https://github.com/michaelfuchs90/ratiobasedcate>. The repository includes the benchmark CSV, the notebook that generates all figures and tables, and a README with replication instructions.

Author contributions. M.F. conceived the project, developed the Q-Learner and the doubly robust augmentations, derived the theoretical results, designed and conducted all empirical experiments, and wrote the manuscript. D.K. engaged in conceptual discussions, and assisted with editing and refining the manuscript.

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A Full Benchmark Results

Figures 3 and 4 present the full per-(learner, dataset) ranking picture as heatmaps: each cell is coloured by $Qini_l(d)/Qini_S(d)$ on a log scale (green = better than S, red = worse), with cells statistically indistinguishable from the S-Learner (paired test, $z = 1.96$) shown in neutral grey. Cells where the learner’s Qini is negative are flagged in deep red (“Qini < 0”), since the ratio breaks down. Tables 3 and 4 below report the underlying mean Qini values (with standard errors).

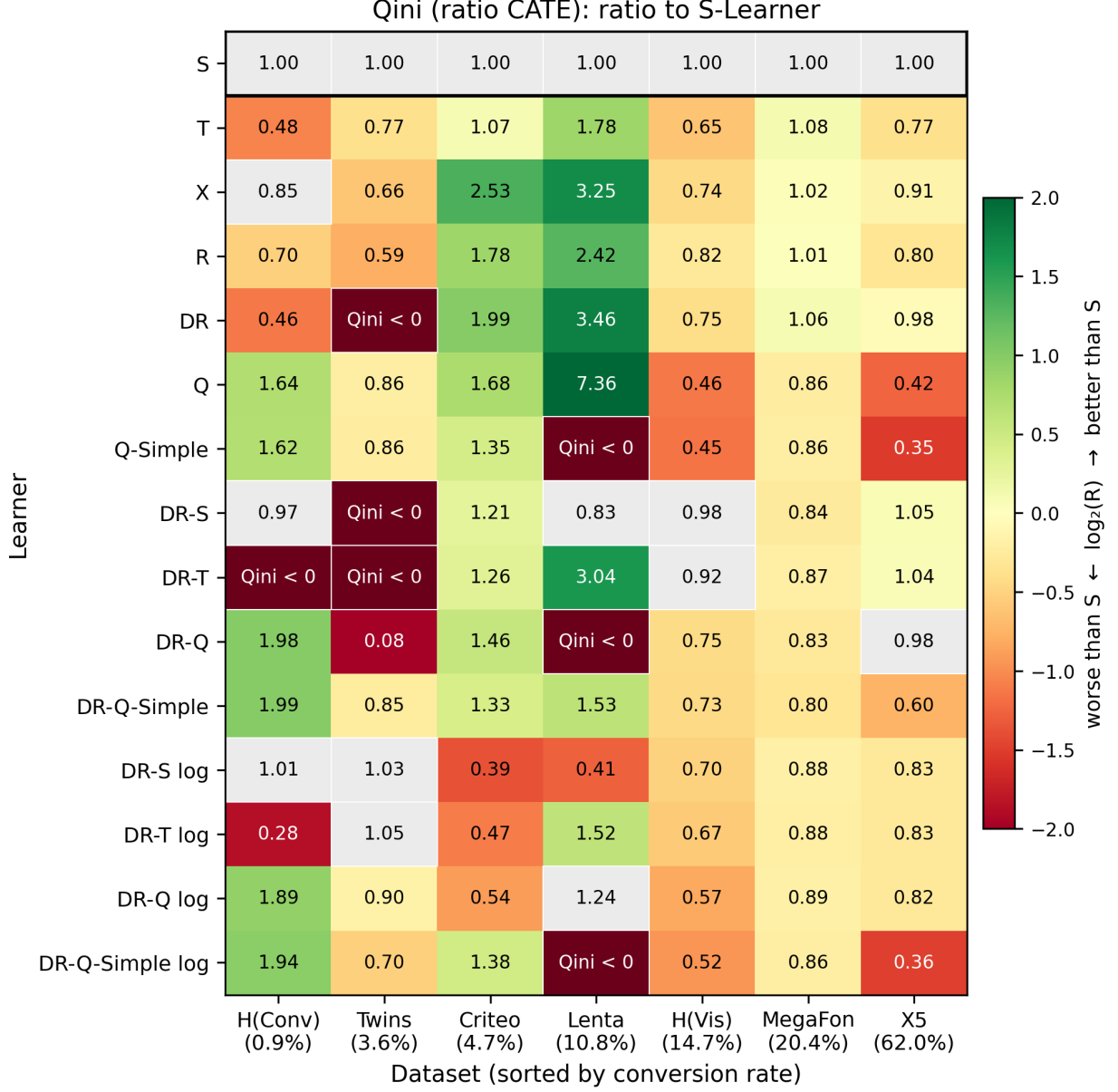


Figure 3: Per-(learner, dataset) heatmap of $Qini_l/Qini_S$ on the seven RCT datasets, sorted by ascending conversion rate. Green cells indicate that the learner beats the S-Learner; red cells indicate the opposite; neutral grey cells are not statistically distinguishable from S at $z = 1.96$.

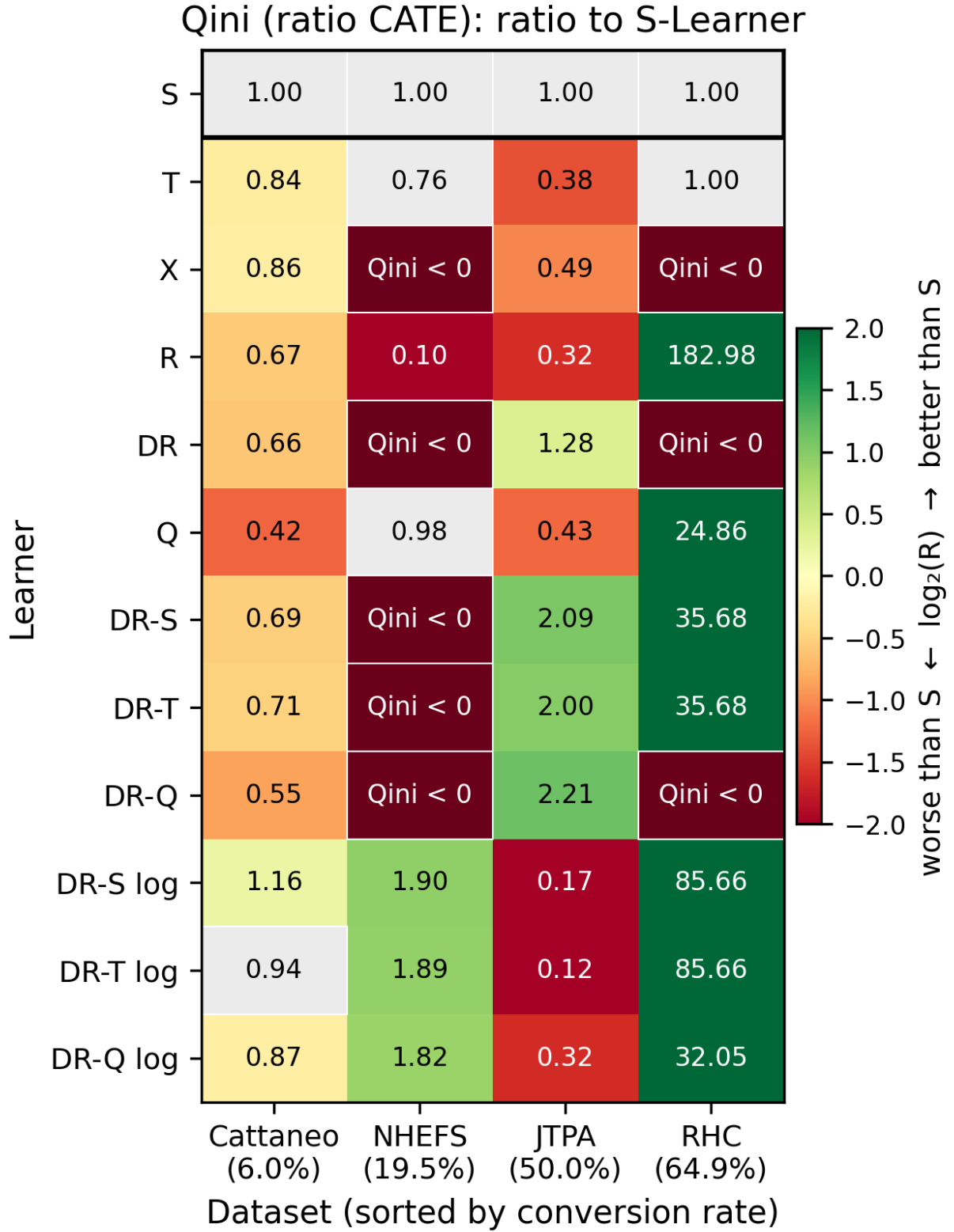


Figure 4: Per-(learner, dataset) heatmap of $Qini_l/Qini_S$ on the four observational datasets, sorted by ascending conversion rate. Conventions as in Figure 3.

Table 3: Qini coefficient (ratio CATE) on RCT datasets (mean over seeds; SE in parentheses). Best per column in **bold**. Bottom row: mean across all learners per dataset.

Learner	H(Conv)	Twins	Criteo	Lenta	H(Vis)	MegaFon	X5
S	0.286 (0.024)	0.087 (0.003)	0.337 (0.003)	0.012 (0.001)	0.437 (0.025)	1.436 (0.007)	0.073 (0.001)
T	0.138 (0.018)	0.067 (0.003)	0.359 (0.003)	0.021 (0.001)	0.285 (0.018)	1.544 (0.007)	0.056 (0.001)
X	0.243 (0.020)	0.057 (0.003)	0.852 (0.006)	0.039 (0.002)	0.322 (0.022)	1.471 (0.007)	0.066 (0.001)
R	0.200 (0.023)	0.051 (0.002)	0.599 (0.006)	0.029 (0.001)	0.357 (0.025)	1.456 (0.008)	0.058 (0.001)
DR	0.133 (0.019)	−0.043 (0.006)	0.671 (0.008)	0.042 (0.002)	0.327 (0.020)	1.524 (0.007)	0.071 (0.001)
Q	0.471 (0.024)	0.075 (0.003)	0.565 (0.005)	0.089 (0.001)	0.202 (0.015)	1.233 (0.006)	0.031 (0.001)
Q-Simple	0.465 (0.023)	0.075 (0.003)	0.454 (0.004)	−0.001 (0.001)	0.196 (0.015)	1.241 (0.006)	0.025 (0.001)
DR-S	0.277 (0.020)	−0.018 (0.004)	0.408 (0.003)	0.010 (0.002)	0.426 (0.023)	1.205 (0.006)	0.077 (0.001)
DR-T	−0.000 (0.015)	−0.010 (0.004)	0.423 (0.004)	0.037 (0.002)	0.404 (0.023)	1.248 (0.005)	0.076 (0.001)
DR-Q	0.568 (0.024)	0.007 (0.004)	0.492 (0.004)	−0.039 (0.002)	0.326 (0.021)	1.193 (0.005)	0.071 (0.001)
DR-Q-Simple	0.571 (0.024)	0.074 (0.002)	0.448 (0.004)	0.018 (0.001)	0.320 (0.020)	1.151 (0.006)	0.044 (0.001)
DR-S log	0.290 (0.021)	0.090 (0.002)	0.130 (0.001)	0.005 (0.001)	0.306 (0.023)	1.267 (0.006)	0.060 (0.001)
DR-T log	0.079 (0.017)	0.092 (0.002)	0.157 (0.002)	0.018 (0.001)	0.294 (0.021)	1.260 (0.006)	0.060 (0.001)
DR-Q log	0.540 (0.023)	0.078 (0.003)	0.181 (0.002)	0.015 (0.001)	0.247 (0.020)	1.272 (0.007)	0.059 (0.001)
DR-Q-Simple log	0.555 (0.025)	0.061 (0.004)	0.465 (0.004)	−0.002 (0.001)	0.227 (0.019)	1.236 (0.007)	0.026 (0.001)
<i>Dataset mean</i>	0.321	0.049	0.436	0.020	0.312	1.316	0.057

Table 4: Qini coefficient (ratio CATE) on observational datasets (mean over seeds; SE in parentheses). Best per column in **bold**. Bottom row: mean across all learners per dataset.

Learner	Cattaneo	NHEFS	JTPA	RHC
S	1.170 (0.062)	0.125 (0.040)	0.032 (0.005)	0.003 (0.002)
T	0.985 (0.052)	0.095 (0.036)	0.012 (0.005)	0.003 (0.002)
X	1.011 (0.060)	-0.095 (0.033)	0.016 (0.005)	-0.444 (0.001)
R	0.789 (0.054)	0.012 (0.028)	0.010 (0.005)	0.483 (0.006)
DR	0.775 (0.051)	-0.027 (0.041)	0.041 (0.006)	-0.086 (0.001)
Q	0.496 (0.054)	0.123 (0.034)	0.014 (0.005)	0.066 (0.003)
DR-S	0.804 (0.050)	-0.048 (0.027)	0.066 (0.006)	0.094 (0.004)
DR-T	0.826 (0.051)	-0.046 (0.032)	0.064 (0.006)	0.094 (0.004)
DR-Q	0.642 (0.053)	-0.105 (0.034)	0.070 (0.006)	-0.104 (0.005)
DR-S log	1.356 (0.066)	0.239 (0.033)	0.006 (0.005)	0.226 (0.007)
DR-T log	1.106 (0.058)	0.237 (0.034)	0.004 (0.005)	0.226 (0.007)
DR-Q log	1.019 (0.065)	0.228 (0.035)	0.010 (0.005)	0.085 (0.005)
<i>Dataset mean</i>	0.915	0.062	0.029	0.054

Table 5: Multiplicative calibration error (ratio CATE) on RCT datasets (mean over seeds; SE in parentheses). Best per column in **bold**.

Learner	H(Conv)	Twins	Criteo	Lenta	H(Vis)	MegaFon	X5
S	2.369 (0.020)	1.403 (0.006)	1.086 (0.001)	1.061 (0.001)	1.227 (0.007)	1.177 (0.002)	1.025 (0.000)
T	3.352 (0.030)	1.955 (0.008)	1.084 (0.003)	1.172 (0.001)	1.494 (0.010)	1.105 (0.002)	1.057 (0.000)
X	3.672 (0.038)	1.993 (0.018)	1.164 (0.004)	1.085 (0.001)	1.400 (0.009)	1.144 (0.002)	1.035 (0.000)
R	2.487 (0.021)	2.064 (0.019)	1.106 (0.003)	1.098 (0.001)	1.435 (0.009)	1.149 (0.002)	1.047 (0.000)
DR	5.624 (0.057)	4.304 (0.029)	2.211 (0.008)	2.688 (0.003)	2.455 (0.020)	2.131 (0.004)	1.665 (0.001)
Q	4.062 (0.043)	2.117 (0.010)	1.115 (0.003)	1.137 (0.001)	1.518 (0.010)	1.130 (0.002)	1.078 (0.000)
Q-Simple	4.061 (0.044)	2.133 (0.010)	1.044 (0.001)	1.252 (0.001)	1.536 (0.010)	1.134 (0.002)	1.075 (0.000)
DR-S	4.505 (0.050)	2.188 (0.013)	1.653 (0.003)	2.047 (0.003)	2.292 (0.020)	2.252 (0.005)	1.334 (0.001)
DR-T	8.174 (0.086)	2.490 (0.017)	1.693 (0.003)	2.149 (0.003)	2.725 (0.023)	2.435 (0.006)	1.343 (0.001)
DR-Q	6.809 (0.090)	2.258 (0.014)	1.718 (0.004)	2.232 (0.003)	2.736 (0.024)	2.227 (0.005)	1.339 (0.001)
DR-Q-Simple	5.278 (0.076)	1.541 (0.006)	1.893 (0.005)	1.901 (0.002)	2.202 (0.019)	1.771 (0.005)	1.535 (0.001)
DR-S log	2.143 (0.017)	1.372 (0.005)	1.139 (0.002)	1.325 (0.002)	1.738 (0.016)	1.167 (0.003)	1.052 (0.000)
DR-T log	2.795 (0.026)	1.562 (0.006)	1.162 (0.002)	1.361 (0.002)	1.918 (0.017)	1.182 (0.003)	1.054 (0.000)
DR-Q log	3.384 (0.035)	1.793 (0.009)	1.153 (0.002)	1.348 (0.002)	1.919 (0.016)	1.135 (0.002)	1.054 (0.000)
DR-Q-Simple log	5.190 (0.055)	2.323 (0.011)	1.073 (0.001)	1.257 (0.001)	1.636 (0.013)	1.118 (0.002)	1.074 (0.000)
<i>Dataset mean</i>	4.260	2.100	1.353	1.541	1.882	1.484	1.184

Table 6: Multiplicative calibration error (ratio CATE) on observational datasets (mean over seeds; SE in parentheses). Best per column in **bold**.

Learner	Cattaneo	NHEFS	JTPA	RHC
S	2.121 (0.023)	1.863 (0.029)	1.195 (0.004)	1.079 (0.002)
T	6.373 (0.104)	3.982 (0.092)	1.366 (0.005)	1.079 (0.002)
X	5.807 (0.112)	3.954 (0.137)	1.314 (0.005)	1.261 (0.006)
R	7.207 (0.187)	4.935 (0.224)	1.395 (0.005)	1.122 (0.003)
DR	11.014 (0.301)	17.125 (0.918)	1.939 (0.008)	1.025 (0.001)
Q	3.809 (0.057)	4.274 (0.088)	1.420 (0.005)	1.772 (0.004)
DR-S	2.872 (0.041)	2.438 (0.040)	1.646 (0.008)	1.129 (0.003)
DR-T	7.198 (0.160)	4.489 (0.113)	1.707 (0.008)	1.129 (0.003)
DR-Q	4.646 (0.077)	3.384 (0.087)	1.711 (0.008)	1.281 (0.005)
DR-S log	3.047 (0.040)	6.254 (0.140)	1.478 (0.006)	1.233 (0.004)
DR-T log	5.758 (0.092)	7.261 (0.203)	1.491 (0.005)	1.233 (0.004)
DR-Q log	4.028 (0.061)	7.858 (0.213)	1.484 (0.005)	1.524 (0.005)
<i>Dataset mean</i>	5.323	5.651	1.512	1.239

B Numerical Stability through Clipping

For numerical stability, we recommend the following clipping bounds as good starting points:

- **Propensity scores:** Clip $\hat{e}(x)$ to $[\epsilon_e, 1 - \epsilon_e]$ with $\epsilon_e = 0.01$
- **Outcome probabilities:** Clip $\hat{\mu}_w(x)$ to $[\epsilon_\mu, 1 - \epsilon_\mu]$ with $\epsilon_\mu = 0.001$
- **Marginal conversion:** Clip $\hat{m}(x)$ to $[\epsilon_m, 1]$ with $\epsilon_m = 0.001$
- **Converter propensity:** Clip $\hat{p}(x)$ to $[\epsilon_p, 1 - \epsilon_p]$ with $\epsilon_p = 0.01$
- **Pseudo-outcomes:** Clip Γ^{DR} to $[\epsilon_\tau, 1/\epsilon_\tau]$ with $\epsilon_\tau = 0.01$
- **Log-scale pseudo-outcomes:** Clip $\Gamma_{\log}^{\text{DR}}$ to $[-C, C]$ with $C = 10$

While these can serve as a good starting point, in practice we recommend treating the clipping parameters as hyperparameters subject to optimization.